MTH601

Quiz no 3

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SOLVED BY

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1. **If the optimum solution of a Linear Programming problem is finitely unique, its corresponding objective function must have:**
   - (a) infinite
   - (b) arbitrary
   - (c) NO
   - (d) finite Correct
2. **While solving a LP by Simplex method, which of the following corresponds to corner points of the feasible region in Graphical method?**
   - (a) non-negative solutions
   - (b) basic feasible solution Correct
   - (c) non-feasible solution
   - (d) non-basic feasible solution
3. **While applying Simplex method to a LP of minimization type, the objective function always:**
   - (a) decreases Correct
   - (b) oscillate
   - (c) increases

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- (d) does not change

4. **While applying Simplex method to a LP of maximization type, the objective function always:**
   - (a) increases Correct
   - (b) oscillates
   - (c) decreases
   - (d) does not change
5. **Shortcoming of Big M method is that the value of M could be:**
   - (a) positive
   - (b) very small
   - (c) negative
   - (d) very large Correct
6. **In the Simplex method, Gauss Jordan Elimination method demands that all the key column's entries should be _____ except key row (pivot) entry:**
   - (a) maximum or minimum
   - (b) strictly negative
   - (c) strictly positive

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- (d) zero Correct

7. **The positive slack variables indicate:**
   - (a) idle(waste) resource
   - (b) excess resource available Correct
   - (c) shortage of resources
   - (d) nil resources
8. **After converting constraints into standard equalities, an LP problem has 4 equations in 6 variables. The number of possible combinations of basic feasible solutions is:**
   - (a) 15 Correct
   - (b) 24
   - (c) 2
   - (d) 10
9. **By Simplex method, to minimize (Z = 2x + 9y), if (z = A > 0) in the initial iteration, the next entering variable is:**
   - (a) y > 0 Correct
   - (b) x < 0
   - (c) x > 0

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- (d) $y < 0$

10. **The first step in the 'Big M' technique is:**

- (a) introducing artificial variables for constraints involving "≥" Correct
- (b) expressing the linear programming problem in equation form
- (c) identifying the objective function coefficients
- (d) None of these

11. **The solution set of the constraint: (2x+3y<12), will lie in:**

- (a) both half planes
- (b) origin side of half plane Correct
- (c) first quadrant only
- (d) opposite to the origin side in half plane

12. **In Simplex Standard table to solve an LP problem, if the ratio (RHS / key column element) is ({2/3,5,1/3,-1/2}), the leaving variable is:**

- (a) Variable corresponding to 5
- (b) Variable corresponding to 1/3 Correct

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- (c) Variable corresponding to -1/2
- (d) Variable corresponding to 2/3

13. **While solving a LP problem by Simplex method, the inequalities are converted into:**

- (a) Equations with negative RHS
- (b) opposite inequalities
- (c) nothing but keep those strictly the same
- (d) Equations with positive RHS Correct

14. **By Simplex method, to minimize (Z = 9x - 2y), if (z = A > 0) in the initial iteration, the next entering variable is:**

- (a) $y < 0$
- (b) $y > 0$
- (c) $x > 0$ Correct
- (d) $x < 0$

15. **Solution region of the constraints (2x+3y>12) and (2x+3y<12) will be:**

- (a) point (6,4)
- (b) empty set Correct

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- (c) straight line (2x+3y=12)
- (d) all first quadrant

16. **While solving a LP problem by Simplex method, the inclusion of slacks in the constraints helps in finding:**

- (a) decision variables
- (b) artificial variables
- (c) non-basic variables
- (d) basic variables Correct

17. **The inequality (3x-2y \ge 14) is equivalent to:**

- (a) (-3x + 2y \le -14) Correct
- (b) (3x-2y < -14)
- (c) (-3x + 2y > 14)
- (d) (3x-2y \le 14)

18. **The straight line (2x+3y<12) meets the x-axis at:**

- (a) 4
- (b) 6 Correct
- (c) 3
- (d) 2

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19. **While solving a LP problem by Simplex, the new basic variable is _____ and the variable removed from the basis is _____:**

- (a) leaving, entering
- (b) feasible, infeasible
- (c) entering, leaving Correct
- (d) infeasible, feasible

20. **While solving a LP problem by Simplex, the element at the intersection of key column and key row is called:**

- (a) Slack
- (b) Leaving
- (c) Entering
- (d) Pivot Correct

21. **In a feasible region, if the end points of the line segment are basic solutions, then all points between these two will:**

- (a) also be basic
- (b) be infeasible
- (c) be non-basic Correct

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- (d) be optimal

22. **The straight line (2x+3y<12) meets the y-axis at:**

- (a) 2
- (b) 4 Correct
- (c) 3
- (d) 6

23. **Simplex method can only be applied to problems with two decision variables:**

- (a) True
- (b) False Correct

24. **After converting constraints into standard equalities, if an initial BFS is (2,4,0,2), then it is:**

- (a) non-degenerate infeasible
- (b) degenerate infeasible
- (c) non-degenerate feasible
- (d) degenerate feasible Correct

25. **In Two-Phase Method, if the minimum value of Phase 1 objective function > 0, then the solution of original problem:**

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- (a) does not exist Correct
- (b) exists
- (c) none of these
- (d) is uncertain

26. **If initial basic solution is _____, no further iteration can be performed:**

- (a) infeasible Correct
- (b) degenerate
- (c) feasible
- (d) non-degenerate

27. **If an artificial variable is not included while converting a constraint of type ≥ into an equation, the solution will be:**

- (a) non-degenerate
- (b) degenerate
- (c) feasible
- (d) infeasible Correct

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28. **In Simplex method for minimization, if at the end of iteration all entries in objective function row are negative, then the problem:**

- (a) cannot be optimized
- (b) has been minimized Correct
- (c) needs further improvement
- (d) has no solution

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